250 Homework #4 (Fall 2026)

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This assignment is due at 11:59 p.m. on Monday, 9 November 2026, to be submitted on Gradescope. Late assignments (with penalty, unless excused) will be accepted until 11:59 p.m. on Tuesday, 10 November 2026, and not after. Please see the syllabus for more details on homework grading and late penalties.

When submitting to Gradescope, please do not forget to assign pages to problems.

The formatting and submission guidelines are the same for this assignment as for the previous ones. Please see the Canvas Homework page for the complete guideline details.

Many of the problems originally come from the textbook and have their original reference numbers, e.g., 4.4.8, attached. Many of those have been modified from the book version. In those cases, you should solve the version given here and not the original one in the book.

Special Induction Rules: Important. Any problem that requires you to prove something by induction should be split into three distinct labelled parts:

(i) State the induction hypothesis $P(x)$ and specify the type of x.
In Q1, this step is already done for you.

(ii) State the base case(s) and explain why they are correct.

(iii) Prove the correctness of the inductive step.
This part must explicitly show exactly where the induction hypothesis is used in the proof.

For reference when doing the problems:
Problems Q1(b), Q2, Q3, Q4, Q5, and Q7(c) require induction.
Problems Q1(c) is best done using induction.
Problems Q1(a), Q6 and Q7 (a) and do not need induction.
Each of the four parts of the EC should be proven by induction, after stating the appropriate Theorem.

*Collaborated with Nobody.

Q1. P4.4.8 [10 pts]

Pig Floyd is weighed at the beginning of every month. In Month 0 he weighs 400 kilograms, in Month 1 he weighs 350 kilograms, and in later months his weight $W(n+1)$ is equal to $\sqrt{2}W(n) - W(n-1) + 700 - 350\sqrt{2}$.

(a) Calculate $W(n)$ for all naturals n with $n \leq 10$.
Write your answers in the form $W(n) = a_n + b_n\sqrt{2}$ where a_n and b_n are integers.

(b) Prove by strong induction on all naturals n that $W(n)$ can be written in the form $a_n + b_n\sqrt{2}$ where a_n and b_n are integers.
For full credit, your induction proof should be split into parts (i), (ii) and (iii) as described on the first page of this document.

(c) Determine $W(84)$, Floyd's weight after seven years (that is, after 84 months). You might find it easiest to discover a pattern in the numbers $W(n)$ from part (a) and then prove that this pattern holds for all n by strong induction.
Reminder. If your proof uses (strong) induction, it should again be split into parts (i), (ii) and (iii) as described n the first page of this document.

Note: For part (a) we only want the values a and b. If you can write a program to find the numbers in the correct $a_n + b_n\sqrt{2}$ format, that's fine. Similarly, if you really want to, you can write a program to solve part (c) but, if you do, you must provide a clean, easily readable documented printout of the program as part of your submission with an explanation of why your program is correct. (The proof might be easier than coding it up).

Solution:

(a)

(i) $W(0) = 400 + 0\sqrt{2}$
(ii) $W(1) = 350 + 0\sqrt{2}$
(iii) $W(2) = 300 + 0\sqrt{2}$
(iv) $W(3) = 350 - 50\sqrt{2}$
(v) $W(4) = 300 + 0\sqrt{2}$
(vi) $W(5) = 350 + 0\sqrt{2}$
(vii) $W(6) = 400 + 0\sqrt{2}$
(viii) $W(7) = 350 + 50\sqrt{2}$
(ix) $W(8) = 400 + 0\sqrt{2}$
(x) $W(9) = 350 + 0\sqrt{2}$
(xi) $W(10) = 300 + 0\sqrt{2}$

(b)

(i) **IH**: Let n be a natural. The induction hypothesis $P(n)$ is the statement that $W(n) = a_n + b_n\sqrt{2}$ where a_n and b_n are integers.

(ii) **BC**: The base cases are $n = 0$ and $n = 1$, since the equation for W depends on the two preceding values of W. For $n = 0$, we know that $W(0) = 400 = 400 + 0\sqrt{2}$, and since 400 and 0 are integers, $P(0)$ is true. For $n = 1$, we know that $W(1) = 350 = 350 + 0\sqrt{2}$, and since 350 and 0 are integers, $P(1)$ is true.

(iii) **IS**: Assume that $P(k)$ is true for all natural numbers k where $0 \leq k \leq n$ and $n \geq 1$. We want to prove $P(n+1)$ is true (that $W(n+1) = a_{n+1} + b_{n+1}\sqrt{2}$ where a_{n+1} and b_{n+1} are integers). By the induction hypothesis, we know that $W(n) = a_n + b_n\sqrt{2}$ and $W(n-1) = a_{n-1} + b_{n-1}\sqrt{2}$, where all the coefficients are integers. Plugging these into the equation for W gives:

$$\begin{aligned} W(n+1) &= \sqrt{2}W(n) - W(n-1) + 700 - 350\sqrt{2} \\ &= \sqrt{2}(a_n + b_n\sqrt{2}) - (a_{n-1} + b_{n-1}\sqrt{2}) + 700 - 350\sqrt{2} \\ &= 2b_n - a_{n-1} + 700 + (a_n - b_{n-1} - 350)\sqrt{2}. \end{aligned}$$

Since $a_n, b_n, a_{n-1}, b_{n-1}$ are all integers, both $2b_n - a_{n-1} + 700$ and $a_n - b_{n-1} - 350$ must also be integers. Therefore, $W(n+1)$ can be expressed in the form $a_{n+1} + b_{n+1}\sqrt{2}$, where a_{n+1} and b_{n+1} are integers. Therefore, $P(n+1)$ is true, which completes the inductive step.

(c) In part a, a pattern can be seen to start repeating at n=8 ($W(0) = W(8), W(1) = W(9), W(2) = W(10)$), which implies that $W(n) = W(n+8)$, which would mean $W(84) = W(84 \mod 8) = W(4) = 300$ (given by part a). Strong Induction Proof:

(i) **IH**: Let n be a natural. The induction hypothesis $P(n)$ is the statement that $W(n) = W(n+8)$.

(ii) **BC**: The base cases are $n = 0$ and $n = 1$, since the equation for W depends on the two preceding values of W. For $n = 0$, we know that $W(0) = 400$ and $W(8) = 400$, so $W(0) = W(8)$, therefore $P(0)$ is true. For $n = 1$, we know that $W(1) = 350$ and $W(9) = 350$, so $W(1) = W(9)$, therefore $P(1)$ is true.

(iii) **IS**: Assume that $P(k)$ is true for all natural numbers k where $0 \leq k \leq n$ and $n \geq 1$. We want to prove $P(n+1)$ is true (that $W(n+9) = W(n+1)$). Plugging these into the equation for W gives:

$$W(n+9) = \sqrt{2}W(n+8) - W(n+7) + 700 - 350\sqrt{2}.$$

By the induction hypothesis, n and $n-1$ must both between 0 and n, so $W(n+8) = W(n)$ and $W(n+7) = W(n-1)$. Plugging these into the equation for W gives:

$$W(n+9) = \sqrt{2}W(n) - W(n-1) + 700 - 350\sqrt{2} = W(n+1).$$

$W(n+9) = W(n+1)$, therefore $P(n+1)$ is true, which completes the inductive step.

Q2. For Section 4.7 [10 pts]

A string w is in B if:
$R_1 : w = \lambda$.
$R_2 : w = avb$, where $v \in B$.
$R_3 : w = uv$, where $u, v \in B$ and $u \neq \lambda, v \neq \lambda$.

Let $N_a(w)$ be the number of a's in string w.
Let $N_b(w)$ be the number of b's in string w.
Let $|w| = N_a(w) + N_b(w)$ be the total number of characters in w.

Prove by induction that, for all strings $w \in B, N_a(w) = N_b(w)$.
That is, the number of 'a's equals the number of 'b's.

We will ask you to prove this in two different ways. The first is by strong induction on the naturals. The second is by structural induction.

For both, we will be checking the correctness of the formulations of your logical statement.

(a) For every natural n, set the inductive hypothesis $P(n)$ to be the statement that "For every string $w \in B$ satisfying $|w| \leq n, N_a(w) = N_b(w)$".
Prove correctness of $P(n)$ for all naturals n in the following two steps:

(ii) What is the base case for induction using $P(n)$?
Show that the base case is correct.

(iii) Prove the correctness of the inductive step that $P(n) \rightarrow P(n+1)$.

(b) For every string $w \in B$, set the inductive hypothesis $P(w)$ to be the statement that "$N_a(w) = N_b(w)$". Prove correctness of $P(w)$ for all strings w in the following two steps:

(ii) What is the base case for induction using $P(w)$?
Show that the base case is correct.

(iii) Given $w \in B$ (and w not the base case), show that $P(w)$ is correct if we assume that every string in B used in the recursive definition of w satisfies $P(w)$.

Solution:

(a) (ii) The base case is $P(0)$ since there is only a single possible string $w \in B$ where $|w| \leq 0$, which is the empty string λ of length 0 (since strings can't have a negative length). By R_1, $\lambda \in B$. Since it is an empty string, it has ZERO 'a's and ZERO 'b's (an equal number of both). Therefore, $P(0)$ is true.

(iii) Assume that $P(n)$ is true. We want to prove that $P(n+1)$ is also true (that for every string $w \in B$ where $|w| \leq n+1$, the number of 'a's equals the number of 'b's). Since $P(n)$ is already true, we only need to consider strings in B where $|w| = n+1$. By the given rules (R_1, R_2, R_3), there are two possible ways to build w (other than R_1, which is unhelpful since it only gives the empty string which can't have a length other than 0).

First, consider R_2 where $w = avb$ and $v \in B$. Since $|w| = n+1$, we know that $|v| = n+1-2$ (-2 for the a and b added to the beginning and end of the string), which is $|v| = n-1$, which is less than n. Therefore, by the induction hypothesis, the number of 'a's and 'b's in v must be equal. Since we add one 'a' to the beginning and one 'b' to the end of v to make w, we're increasing the number of 'a's and 'b's in the string by an equal amount, therefore the number of 'a's and 'b's in w is still equal.

Next, consider R_3 where $w = uv$ and $u, v \in B$, with $u \neq \lambda$ and $v \neq \lambda$. Since both u and v are non-empty, their lengths must be greater than 0 and less than $|w| = n+1$, meaning that $|u| \leq n$ and $|v| \leq n$. Therefore, by the induction hypothesis, the number of 'a's and 'b's must be equal in both u as well as v. Since w is formed by concatenating u and v, we have to add their number of 'a's and 'b's together to get the total number of 'a's and 'b's in w; and since the number of 'a's and 'b's are equal for both strings, the sum of their 'a's and the sum of their 'b's must also be equal.

Therefore, in both possible cases, the number of 'a's and 'b's is equal, making $P(n+1)$ true, completing the inductive step.

(b) (ii) The base case is $w = \lambda$ since R_1 is the rule that constructs the empty string. By R_1, $\lambda \in B$. Since it is an empty string, it has ZERO 'a's and ZERO 'b's. Therefore, $N_a(\lambda) = N_b(\lambda) = 0$, and $P(\lambda)$ is true.

(iii) Let $w \in B$ be any string that's not the base case (λ). We want to prove that $P(w)$ is true; that is, we want to show that $N_a(w) = N_b(w)$. Since w is not the base case, it must have been made using either R_2 or R_3.

First, consider R_2, where $w = avb$ and $v \in B$. By the induction hypothesis, we know that $P(v)$ is true, meaning that $N_a(v) = N_b(v)$. Since we add one 'a' to the beginning and one 'b' to the end of v to make w, we're increasing the number of 'a's and 'b's in the string by an equal amount, therefore the number of 'a's and 'b's in w is still equal.

Next, consider R_3, where $w = uv$ and $u, v \in B$, with $u \neq \lambda$ and $v \neq \lambda$. By the induction hypothesis, we know that $P(u)$ and $P(v)$ are both true, meaning that $N_a(u) = N_b(u)$ and $N_a(v) = N_b(v)$. Since w is formed by concatenating u and v, we have to add their number of 'a's and 'b's together to get the total number of 'a's and 'b's in w; and since the number of 'a's and 'b's are equal for both strings, the sum of their 'a's and the sum of their 'b's must also be equal.

Therefore, in both possible cases, $N_a(w) = N_b(w)$, making $P(w)$ true for all $w \in B$, completing the inductive step.

Q3. P4.6.5 [10 pts]

Define the *exponentiation operator* on naturals recursively so that $x^0 = 1$ and $x^{S(n)} = x^n \cdot x$. Using this definition:

(a) Prove by induction that for any naturals x, y, and z, $x^{y+z} = x^y \cdot x^z$

(b) Prove by induction that for any naturals x, y, and z, $x^{y \cdot z} = (x^y)^z$.

Hints:

1. Don't forget to split the induction proof into the three parts, (i), (ii) and (iii).
2. To prove (b), try replacing x in (a) by the natural x^y.
3. When trying to prove something is true for three variables, our standard approach is to arbitrarily fix two of the variables and do induction on the third. For example, in (a), you should fix x, y and try to prove by induction on z, the proposition $P_{x,y}(z)$, which is is that "$x^{y+z} = x^y \cdot x^z$"
4. You may use any of the known properties of addition and multiplication that we have taught.
5. As an example, for part (a) (iii) you might start with the assumption $P_{x,y}(z)$ and show that it implies $P_{x,y}(S(z))$, where $S(z)$ is the successor of z. So your proof structure might look something like

$$\begin{aligned} x^{y+S(z)} &= x^{S(y+z)} && \text{Definition of } S(n) \\ &= \dots \\ &\vdots \\ &= x^y \cdot x^{S(z)}. \end{aligned}$$

Solution:

(a) (i) **IH**: Induction on z, let x and y be arbitrary naturals and let $P_{x,y}(z)$ be the proposition: $x^{y+z} = x^y \cdot x^z$.

(ii) **BC**: The base case is $z = 0$, which tries to prove $x^{y+0} = x^y \cdot x^0$. Since $y + 0 = y$ and $x^0 = 1$, $x^y = x^y \cdot 1 = x^y$. Therefore, $P_{x,y}(0)$ is true.

(iii) **IS**: Assume that $P_{x,y}(z)$ is true for some natural z. (that $x^{y+z} = x^y \cdot x^z$). We want to prove that $P_{x,y}(S(z))$ is also true: $x^{y+S(z)} = x^y \cdot x^{S(z)}$.

$$\begin{aligned}
x^{y+S(z)} &= x^{S(y+z)} && \\
&= x^{y+z} \cdot x && \text{induction hypothesis} \\
&= (x^y \cdot x^z) \cdot x && \text{exponent product rule} \\
&= x^y \cdot (x^z \cdot x) && \text{associative property} \\
&= x^y \cdot x^{z+1} && \text{exponent product rule} \\
&= x^y \cdot x^{S(z)} && \text{definition of successor}
\end{aligned}$$

Therefore, $P_{x,y}(S(z))$ is true, completing the inductive step. Thus $x^{y+z} = x^y \cdot x^z$ for all natural numbers x, y, and z.

(b) (i) **IH**: Induction on z, let x and y be arbitrary naturals and let $P_{x,y}(z)$ be the proposition: $x^{y \cdot z} = (x^y)^z$.

(ii) **BC**: The base case is $z = 0$, which tries to prove $x^{y \cdot 0} = (x^y)^0$. Since $y \cdot 0 = 0$ and anything raised to the power of zero is 1, $x^0 = (x^y)^0 = 1$. Therefore, $P_{x,y}(0)$ is true.

(iii) **IS**: Assume that $P_{x,y}(z)$ is true for some natural z. (that $x^{y \cdot z} = (x^y)^z$). We want to prove that $P_{x,y}(S(z))$ is also true: $x^{y \cdot S(z)} = (x^y)^{S(z)}$.

$$\begin{aligned}
(x^y)^{S(z)} &= (x^y)^z \cdot x^y && \\
&= x^{y \cdot z} \cdot x^y && \text{induction hypothesis} \\
&= x^{y \cdot z + y} && \text{part (a)} \\
&= x^{y(z+1)} && \text{distributive property} \\
&= x^{y \cdot S(z)} && \text{definition of successor}
\end{aligned}$$

Therefore, $P_{x,y}(S(z))$ is true, completing the inductive step. Thus $x^{y \cdot z} = (x^y)^z$ for all naturals x, y, z.

Q4. Modification of P4.7.1 [10 pts]

Let w be a string over some alphabet Σ. Recall that the reversal operator w^R is defined recursively by (i) $\lambda^R = \lambda$ and, for every string w, and character $a \in \Sigma$, $(wa)^R = aw^R$.

(a) Prove by induction on strings that, for every string w, and character $a \in \Sigma$, $(aw)^R = w^R a$.

Hint. The end of Lecture 19 proves that for all strings, u, v, $(wu)^R = u^R w^R$. *You can modify that proof to solve this*

(b) Prove by induction on strings that for any string $w, (w^R)^R = w$.
Hint. Use part (a).

Solution:

(a) (i) Induction on strings w over the alphabet Σ. Let $P(w)$ be $(aw)^R = w^R a$, where a is some arbitrary character in Σ.

(ii) The base case is $w = \lambda$ which tries to prove $(a\lambda)^R = \lambda^R a$. Since $a\lambda = a$ and $\lambda^R = \lambda$, we have: $a^R = a$. By the definition of reversal recursively, $(\lambda a)^R = a\lambda^R = a$. Therefore: $(a\lambda)^R = a = \lambda a = \lambda^R a$. Thus, $P(\lambda)$ is true.

(iii) Let w be any string other than the base case (λ) so that it isn't empty, and assume that $P(v)$ is true for the string v when $w = vb$, where b is a character in Σ. The goal is to prove $(avb)^R = (vb)^R a$. Substituting $w = vb$ into the original equation $(aw)^R = (w)^R a$, gives:

$$\begin{aligned} (avb)^R &= b(av)^R && \text{def of reversal recursively} \\ &= b(v^R a) && \text{induction hypothesis} \\ &= (vb)^R a. && \text{def of reversal recursively} \end{aligned}$$

Therefore, $P(w)$ is true, completing the inductive step.

(b) (i) Induction on strings w over the alphabet Σ. Let $P(w)$ be $(w^R)^R = w$.

(ii) The base case is $w = \lambda$ which tries to prove $(\lambda^R)^R = \lambda$. By the definition of reversal recursively, $\lambda^R = \lambda$. Therefore $(\lambda^R)^R = \lambda^R = \lambda$. Thus, $P(\lambda)$ is true.

(iii) Let w be any string other than the base case (λ) so that it isn't empty, and assume that $P(v)$ is true for the string v when $w = vb$, where b is a character in Σ. The goal is to prove $((vb)^R)^R = vb$.

$$\begin{aligned} ((vb)^R)^R &= (bv^R)^R && \text{def of reversal recursively} \\ &= (v^R)^R b && \text{part (a)} \\ &= vb. && \text{induction hypothesis} \end{aligned}$$

Therefore, $P(w)$ is true, completing the inductive step.

Q5. Modified P4.11.1 [10 pts]

Prove that a $3 \times n$ rectangle can be covered exactly with L-shaped tiles if and only if n is even. Do this in two parts:

(a) Prove by induction that if n is even, then the board can be tiled. More explicitly prove that for all natural n, a $3 \times (2n)$ can be tiled exactly with L-shaped tiles.

(b) Prove by induction that if n is odd, then the board cannot be tiled. More explicitly prove that for all natural n, a $3 \times (2n+1)$ board cannot be tiled exactly with L-shaped tiles.

***Hint**: For (b) it it might help to use an indirect proof, i.e., proof by contradiction. For example, let $Q(n)$ be the proposition that a $3 \times (2n+1)$ board cannot be tiled exactly with L-shaped tiles. Then, in the inductive step, after assuming $Q(n)$, instead of proving directly that $Q(n) \to Q(n+1)$, prove instead the equivalent $\neg Q(n+1) \to \neg Q(n)$.*

Solution:

(a) (i) **IH**: Let n be a natural and $P(n)$ be the proposition that a $3 \times (2n)$ rectangle can be tiled exactly with L-shaped tiles.

(ii) **BC**: The base case is $n = 0$. A 3×0 rectangle is empty, so it can be trivially be tiled using zero L-shaped tiles. Therefore $P(0)$ is true.

(iii) **IS**: Assume that $P(n)$ is true for some natural n (that a $3 \times (2n)$ rectangle can be tiled exactly with L-shaped tiles). The goal is to prove that $P(n+1)$ is also true (that a $3 \times (2(n+1))$ rectangle can be tiled exactly with L-shaped tiles). By the induction hypothesis, the first $2n$ columns can already be tiled. We can extend this tiling by adding two more columns, which form a 3×2 rectangle which can then be tiled using two L-shaped tiles with the corners of each L-shaped tile in any two opposite corners of the rectangle (top left and bottom right, or top right and bottom left). Therefore $P(n+1)$ is true, completing the inductive step. By induction, every $3 \times (2n)$ rectangle can be tiled exactly with L-shaped tiles.

(b) (i) **IH**: let n be a natural and $Q(n)$ be the proposition that a $3 \times (2n+1)$ rectangle ***can't*** be tiled exactly with L-shaped tiles.

(ii) **BC**: The base case is $n = 0$. A 3×1 rectangle cannot be tiled with L-shaped tiles because an L-shaped tile requires a 2×2 area at least, but the rectangle has only one column. Therefore $Q(0)$ is true.

(iii) **IS**: Assume that $Q(n)$ is true for some natural n (that a $3 \times (2n+1)$ rectangle can't be tiled exactly with L-shaped tiles). The goal is to prove that $Q(n+1)$ is also true (that a $3 \times (2(n+1)+1)$ rectangle cannot be tiled exactly with L-shaped tiles). The first $2n$ columns can already be tiled. We can extend this tiling by adding two more columns, which form a 3×2 rectangle which can then be tiled

using two L-shaped tiles with the corners of each L-shaped tile in any two opposite corners of the rectangle (top left and bottom right, or top right and bottom left); this leaves a 3×1 rectangle, which cannot be tiled with L-shaped tiles because an L-shaped tile requires a 2×2 area at least, but the rectangle has only one column. Therefore $Q(n+1)$ is true, completing the inductive step. By induction, every $3 \times (2n+1)$ rectangle can't be tiled exactly with L-shaped tiles.

Q6. P9.1.4 [10 pts]

Let G be a tree and let x and y be any two nodes such that (x, y) is not an edge. Let H be the graph obtained from G by adding the edge (x, y). Prove that H has exactly one simple cycle.

Hint: You may use the Unique Path Theorem, taught in class and page 9-3 in the textbook.

Solution:

Let G be a tree, and let (x, y) be any two nodes such that (x, y) is not an edge in G. Let H be the graph obtained by adding the edge (x, y) to G. Since G is a tree, all the nodes are connected in some way and it has no cycles. By the Unique Path Theorem, there is exactly one simple path from y to x in G. Let this path be $y, v_0, v_1, \ldots, v_k, x$. Since H contains all the edges of G, as well as the newly added edge (x,y), the path from x to y can be added to the end of the path established earlier in G: $y, v_0, v_1, \ldots, v_k, x, y$, which forms a simple cycle. There can't be any more than exactly 1 cycle, since only the 1 edge ((x, y)) was added to G to make H, and multiple cycles can't be created by adding just 1 edge unless there were pre-existing cycles (which G doesn't have). Therefore H has exactly one simple cycle.

Q7. P9.3.2 [10 pts]

Recall the definition of the Fibonacci numbers

$$F(n) = \begin{cases} 0 & \text{if } n = 0 \\ 1 & \text{if } n = 1 \\ F(n-1) + F(n-2) & \text{if } n > 1 \end{cases}. \tag{1}$$

Exercise 9.3.2 defines the following Pseudo-Java method that correctly computes the Fibonacci numbers $F(n)$ on any **natural** input n.

```
natural fib (natural n) {
     if (n <= 1)
        return n;
     return fib(n-1) + fib (n-2);
}
```

Let $G(n)$ denote the total number of recursive calls are made by the method when called on initial value n. The initial values are $G(0) = G(1) = 0$ and $G(2) = 2$.

(a) Find an equation describing $G(n)$ by replacing the question marks in the recurrence below with appropriate numbers. Explain how you derived this from the code.

$$G(n) = \begin{cases} ?? & \text{if } n = 0 \\ ?? & \text{if } n = 1 \\ G(n-1) + G(n-2) + ?? & \text{if } n > 1 \end{cases}. \tag{2}$$

(b) Calculate the values of $G(n)$ for all $n \leq 8$.

(c) Prove by induction that, for all n, $G(n) = 2F(n+1) - 2$.

Solution:

(a)

$$G(n) = \begin{cases} 0 & \text{if } n = 0 \\ 0 & \text{if } n = 1 \\ G(n-1) + G(n-2) + 2 & \text{if } n > 1 \end{cases}. \tag{3}$$

Because when $n \leq 1$, the method returns n without making any recursive calls, and when $n > 1$, the method calls fib(n-1) and fib(n-2), making two recursive calls in addition to all the recursive calls made by those two methods. Therefore, we add $G(n-1)$, $G(n-2)$, plus the initial 2.

(b)

- $G(0) = 0$

- $G(1) = 0$
- $G(2) = 2$
- $G(3) = 4$
- $G(4) = 8$
- $G(5) = 14$
- $G(6) = 24$
- $G(7) = 40$
- $G(8) = 66$

(c) (i) **IH**: Let $P(n)$ be the proposition that $G(n) = 2F(n+1) - 2$ where n is a natural.

(ii) **BC**: The base cases are $P(0)$ and $P(1)$. For n=0 $G(0) = 0 = 2F(1) - 2 = 2(1) - 2 = 0$, Therefore P(0) is true. For n=1 $G(1) = 0 = 2F(2) - 2 = 2(1) - 2 = 0$, Therefore $P(1)$ is true.

(iii) **IS**: Assume that $P(k)$ is true for every natural where $k < n$ and $n \geq 2$. The goal is to prove $G(n) = 2F(n+1) - 2$ by the recurrence for $G(n)$,

$$\begin{aligned} G(n) &= G(n-1) + G(n-2) + 2 \\ &= (2F(n) - 2) + (2F(n-1) - 2) + 2 && \text{induction hypothesis} \\ &= 2F(n) + 2F(n-1) - 2 \\ &= 2\left(F(n) + F(n-1)\right) - 2 \\ &= 2F(n+1) - 2 && \text{def of Fibonacci recursively} \end{aligned}$$

Therefore, $P(n)$ is true, completing the inductive step.

Extra Credit: P9.1.5 [10 pts]

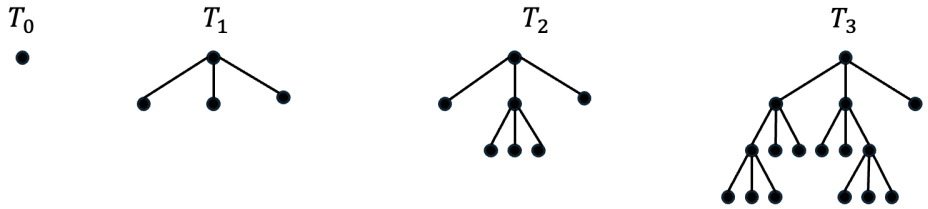
Definition: *A rooted tree has degree k if every internal node has k children. $\mathcal{T}$ will denote the set of all directed trees of degree k.*

For the purpose of this problem $\mathcal{T}$ is defined recursively as follows:

- The one node tree is in $\mathcal{T}$.
- Tree T is in $\mathcal{T}$ if T has a root with k children and each of its k children is a tree in $\mathcal{T}$.
- A tree is in $\mathcal{T}$ only if it can be derived using the first two rules.

More Notation: use $d(T)$ to denote the depth of T, $L(T)$ to denote the number of leaves in T and $|T|$ to denote the total number of nodes (leaves plus internal nodes) in T.

The diagram shows four 3-regular rooted trees with their associated depths, number of nodes and leaves. Note that the root of T_1 has three children, all T_0 trees. The root of T_2 has three children, the first and third are T_0 trees and the second is a T_1 tree.



(a) State and prove a theorem giving the maximum possible number of leaves in a rooted tree of depth d and degree k.

(b) State and prove a theorem giving the minimum possible number of leaves in a rooted tree of depth d and degree k.

(c) State and prove a theorem giving the maximum possible number of nodes in a rooted tree of depth d and degree k.

(d) State and prove a theorem giving the minimum possible number of nodes in a rooted tree of depth d and degree k.

Note: each of these proofs should be done in two parts. In the first part, a bound should be proven by induction. In the second, the bound should be proven tight.

For example, in (a) you should find a function $f(d, k)$ and prove by induction that the number of leaves in a rooted tree of depth d and degree k is at most $f(d, k)$. An induction proof is necessary here since the trees are defined recursively. You should then, for each d, k, describe a tree that meets those bounds.

Solution:

(a) The max leaves in a rooted tree of depth d and degree k is $L(T) = k^d$

(b) The min leaves in a rooted tree of depth d and degree k is $L(T) = (k-1)d + 1$

(c) $1 + k + k^2 + \cdots + k^d$

(d) $kd + 1$